

O'RUME DOMINIC URIRIE

AI Engineer · Agentic AI Assurance | LLM and RAG systems, verifiable accountability, measurement instrumentation

CONTACT

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SKILLS

Languages & Frameworks

Python (production-grade), TypeScript / JavaScript, SQL, FastAPI, Flask, Next.js, React

AI & Agentic Systems

LLM orchestration, RAG pipelines, agent tool-use and planning, human-in-the-loop control, LangChain, LangGraph, CrewAI, local LLMs (Ollama), LLM evaluation

Machine Learning & Research Method

scikit-learn, pandas, PyTorch, classification and forecasting, anomaly detection, pre-registered experimental design, blinded analysis, inter-rater reliability (Cohen's κ), ANOVA

Cloud, Delivery & Data

Docker, GitHub Actions, GitLab CI, Google Cloud Run, Vercel, Fly.io, AWS, Supabase, Elastic Stack, OCR pipelines, PyPI packaging, automated testing

Verification & Governance

SHA-256 audit chains, on-chain anchoring (Polygon), Daml smart contracts, STRIDE threat modelling, RBAC, secrets management

CERTIFICATIONS

Microsoft Azure AI Apps & Agents Developer
Certified Blockchain Architect, Blockchain Council
AI Governance & Responsible AI
Leveraging AI for Governance, Regulation and Compliance
Python, University of Michigan
Cisco CCNA and CCNP

PROFILE

AI engineer working where agentic systems meet assurance. I build the instrumentation that answers a question conventional benchmarks do not ask: not whether generated code runs, but whether the agent built the thing it was asked to build. My MSc instrument is open source, pre-registered and published with its own errata; the companion ledger work is catalogued in the Canton Network's official developer hub. Alongside the research I ship production systems: LLM and RAG services, document-verification layers and multi-tenant SaaS, deployed and owned end to end.

PROFESSIONAL EXPERIENCE

MCKI Solutions

AI Engineer, Enterprise Applications

November 2025 to present · Birmingham, UK

- Architect and deploy enterprise AI applications from raw requirements through to cloud-hosted production systems, across student routing, document processing, content automation and housing operations.
- Built an OCR and RAG pipeline at 98% precision that extracts transactions, classifies tax categories and accelerates compliance review.
- Developed an AI student gateway mapping applicant preferences to universities and visa pathways, cutting manual support queries by 40%.
- Own cloud deployment, performance budgets and the CI gates every change must pass, holding a consistent 90+ Lighthouse score.

VOREM

AI / ML Engineer and Technical Lead

September 2017 to October 2025

- Moved with the field across two eras: classical ML pipelines and predictive modelling from 2017 (scikit-learn, pandas), then LLM and agentic systems from 2023.
- Engineered a five-model LLM analytics workflow on Google Cloud Run, owning uptime, latency and cost.
- Built a supervised-autonomy quantitative FX and crypto signal system on a nine-module architecture, with API delivery channels and a governance kill-switch.
- Technical project supervisor: led development and testing of AI tools across prompt engineering, generative AI and API integration.
- Monitored production systems using Elastic observability for anomaly detection, and reviewed IT security processes against the findings.

Simera Global

AI Advisor (advisory engagement)

February 2026 to present · London, UK

- Advise on the technical design of AI workflows: CrewAI and LangGraph orchestration, verification layers, and secure cross-border data handling for autonomous-agent use cases.

EDUCATION

Aston University: MSc Artificial Intelligence with Business Strategy

2026 · Birmingham, UK

Dissertation: *Measuring the Unmeasured*, a pre-registered, blinded instrument for auditing the code-quality and governance behaviour of agentic AI coding workflows. Aligned to the Aston and Capgemini Centre of Excellence for Enterprise AI. ML coursework on UK road-accident and cardiovascular data for classification and predictive modelling.

Covenant University: BEng Electrical & Electronic Engineering

2014

Final project: a 5 kVA intelligent changeover inverter. Industrial internship at a 1,020 MW thermal power plant.

SELECTED PROJECTS

AI Code Quality Auditor: open-source measurement instrument

Python · published on PyPI · github.com/dominicrume/ai-code-quality-auditor

An empirical safety harness for agentic AI coding systems. It scores five metrics (security-vulnerability density, cyclomatic complexity, duplication, specification hallucination and correction frequency) across five workflow conditions against fixed specifications, over 600 controlled measurements, with the analysers blinded by interface design rather than by discipline. Published on PyPI with over 3,600 downloads, shipped as a reusable GitHub Action with a CI gate and a JSON decision record, and served by a live public dashboard.

Know Your AgenticAI: spend mandates and hash-chained receipts on Canton

Daml · Python · catalogued in the official Canton Developer Hub

Enforces an AI agent's spending mandate inside a Daml choice body (cap, per-period limit, allow-list, expiry, revocation) and seals every attempt into a hash-chained receipt, including the ones the ledger refused, so refusals reach an auditor without exposing the accepted payments behind them. **Three pull requests merged by the Canton Foundation's developer-relations team**, catalogued as a partner tool in the AI Tools category.

Tenant Hub: enterprise housing-operations SaaS

Next.js · Supabase · MCKI Solutions · live April 2026

Multi-tenant platform serving 100+ tenants across housing brands, with role-based access control, brand-level data separation, SHA-256 audit-chain records and CI-enforced deployment controls, so every action is attributable and tamper-evident. The automation saves the operator roughly 100 hours per week and about £80,000 per year.

VeritaPort: evidentiary audit layer for a legal-tech platform

FastAPI · Polygon · HMAC-SHA256 · client engagement, integrated by a London disputes practice

Built the cryptographic audit-trail layer for an institutional decision-capture platform: live Polygon audit stamping with document tamper verification, so any third party can verify record integrity in seconds without per-user wallets. Followed it with the enterprise API layer, adding scoped API keys, per-firm multi-tenant isolation, HMAC-SHA256 signed webhooks and an OpenAPI specification. Chose 404 over 403 on cross-tenant reads so a firm cannot learn that another firm's matter exists.

Local 7B LLM email processor

Ollama · runs entirely on-device

Evaluates incoming email locally, deletes spam, leaves normal messages unread and flags receipts for a paperless pipeline. Built as a daily driver, with no data leaving the machine.

AI job search agent: autonomy under review

Claude API · Playwright · Flask

Discovers roles, compares CV fit and drafts tailored applications, routing every output through a human-approval dashboard before submission. Browser automation and tool use under explicit human-in-the-loop governance.

RESEARCH, IP & RECOGNITION

- **UK patent application GB2611754.9**, filed with the UK IPO on 20 May 2026. Pending and not yet published; filing receipt available on request. Covers the KYA framework for verifiable AI accountability and provenance.
- **Named technical partner on an Innovate UK Frontier AI consortium bid** with Learncast and the University of Birmingham, with the patent-pending KYA framework as background IP. Ownership and terms confirmed in writing by the commercial lead and by the University's Impact and Engagement Manager, School of Computer Science.
- **ACM Europe Seasonal School on Responsible AI**, Aston University, September 2026. Selected through a scored international application round that invited only the top 30–50 candidates. My team was awarded second prize in the school's project competition, for human-in-the-loop decision support in regulated markets.
- **Unsolicited security review of OpenZeppelin's Canton contracts**. Filed three issues showing that authorisation fences ship without tests that fail when the fence is removed: 25 of 30 in *canton-stablecoin* including the oracle price guard, 24 of 32 in *canton-token-template*, and three access-control fences in *canton-contracts*.
- **FrontierTechX**, University of Birmingham, March 2026. Panellist, "Agentic AI: From Copilots to Autonomous Enterprises". Also hosted and spoke at two further Birmingham technology events in 2026.
- **MSc dissertation** validated by two independent raters (Cohen's $\kappa = 0.870$ rater-to-rater), with the erratum that overturned one of its own cleanest findings published rather than quietly corrected.
- **Author of five books** on AI and blockchain, listed and orderable through Waterstones and Amazon. Foreword to *From Code to Consciousness* by Olusola Amusan, co-founder of Vesti.

REFERENCES

Contact details for both are available on request.

MCKI Solutions

Employer, November 2025 to present

VOREM

Employer, 2017 to 2025